

LEGO 2 → AULA 6, VARIÂNCIA DOS ESTIMADORES

EXPECTED VALUE

$X_1, X_2, \dots, X_N \leftarrow \text{population}$

the population mean is $E[X] = \mu = \frac{X_1 + X_2 + \dots + X_N}{N}$

$$E[2X] = \frac{2X_1 + 2X_2 + \dots + 2X_N}{N} = 2 \cdot E[X]$$

$$\Rightarrow E[aX] = a \cdot E[X]$$

Now suppose X is income and Y is bonus:

$$E[X] = \mu_x = \frac{X_1 + X_2 + \dots + X_N}{N}, \quad E[Y] = \mu_y = \frac{Y_1 + Y_2 + \dots + Y_N}{N}$$

Total money:

$$E[X+Y] = \frac{X_1+Y_1 + X_2+Y_2 + \dots + X_N+Y_N}{N} = \frac{X_1 + \dots + X_N}{N} + \frac{Y_1 + \dots + Y_N}{N}$$

$$= E[X] + E[Y]$$

$$\text{Linearity of Expectation } E[2X+3Y] = E[2X] + E[3Y]$$

$$= 2E[X] + 3E[Y]$$

$$= aE[X] + bE[Y]$$

VARIANCE

$$\text{Var}(X) = \sigma_x^2 = \frac{(X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_N - \mu)^2}{N}$$

$$\text{Var}(2X) = \frac{(2X_1 - 2\mu)^2 + (2X_2 - 2\mu)^2 + \dots + (2X_N - 2\mu)^2}{N}$$

$$= \frac{4(X_1 - \mu)^2 + 4(X_2 - \mu)^2 + \dots + 4(X_N - \mu)^2}{N}$$

$$= 4 \text{Var}(X) \Rightarrow \text{Var}(aX) = a^2 \text{Var}(X)$$

VARIANCE AS AN EXPECTATION

$$\text{Var}(X) = \sigma^2 = \frac{(X_1 - \mu)^2 + (X_2 - \mu)^2 + \dots + (X_N - \mu)^2}{N}$$

$$= E[(X - \mu)^2] = E[(X - \mu)(X - \mu)]$$

$$= E[X^2 - 2\mu X + \mu^2] = E[X^2] + E[-2\mu X] + E[\mu^2]$$

$$= E[X^2] - 2\mu E[X] + \mu^2$$

$$= E[X^2] - 2\mu^2 + \mu^2$$

$$= E[X^2] - \mu^2 \Rightarrow E[X^2] - (E[X])^2$$

$$\text{Var}(X) = E[X^2] - \mu^2$$

VARIANCE AND COVARIANCE

$\text{Var}(aX) = a^2 \text{Var} X$. But what about $\text{Var}(X+Y)$?

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

se a covariância for positiva, maior a variância
negativa, menor " " "

se X e Y são independentes, $\text{Cov}(X, Y) = 0$

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$\text{Var}(X+Y) = E[(X+Y - (E[X] + E[Y]))^2]$$

$$= E[(X+Y - E[X] - E[Y])^2]$$

$$= E[(X - E[X]) + (Y - E[Y])^2]$$

$$= E[(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])]$$

$$= \underbrace{E[(X - E[X])^2]}_{\text{Var } X} + \underbrace{E[(Y - E[Y])^2]}_{\text{Var } Y} + 2 \underbrace{E[(X - E[X])(Y - E[Y])]}_{\text{Cov}(X, Y)}$$

$$\text{Var}(aX+bY) = a^2 \text{Var} X + b^2 \text{Var} Y + 2ab \text{Cov}(X, Y)$$

S^2 is unbiased

$$E[(X - \mu)^2] = E[X^2] - \mu^2$$

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2)$$

$$= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2$$

$$= \sum_{i=1}^n x_i^2 - 2\bar{x} \cdot n\bar{x} + n \cdot \bar{x}^2$$

$$= \sum_{i=1}^n x_i^2 - 2\bar{x}^2 n + n \cdot \bar{x}^2$$

$$= \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

So...

At the population level: $E[(X - \mu)^2] = E[X^2] - \mu^2$

At the sample level: $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2) - n\bar{x}^2$

To show that s^2 is unbiased for σ^2 , we need to show:

$$E[s^2] = \sigma^2$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right] = E\left[\sum_{i=1}^n x_i^2 - n\bar{x}^2\right]$$

$$= E\left[\sum_{i=1}^n x_i^2\right] - nE[\bar{x}^2] = \sum_{i=1}^n E[x_i^2] - nE[\bar{x}^2]$$

$$= \sum_{i=1}^n (\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2 = \sigma^2(n-1)$$

$$\text{Var}(X_i) = E[X_i^2] - (E[X_i])^2$$

$$E[X_i^2] = \text{Var}(X_i) + (E[X_i])^2$$

$$\Rightarrow E[s^2] = E\left[\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2\right]$$

$$= \frac{1}{n-1} E\left[\sum_{i=1}^n (x_i - \bar{x})^2\right]$$

$$= \frac{1}{n-1} \sigma^2 (n-1) = \sigma^2$$

EXPECTATION AND VARIANCE OF A RANDOM VECTOR

EXPECTATION

$$\underline{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{pmatrix} \quad E[\underline{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix} \Rightarrow E[X_1 + X_2] = E\begin{bmatrix} X_{11} + X_{21} \\ X_{12} + X_{22} \\ \vdots \\ X_{1N} + X_{2N} \end{bmatrix}$$

$$E[X_1 + X_2] = E[X_1] + E[X_2]$$

$$E[A\underline{X}] = A \cdot E[\underline{X}] \rightsquigarrow E[aX] = aE[X]$$

VARIANCE OF A RANDOM VECTOR

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} \quad \text{we could think the variance would be...} \quad \begin{bmatrix} \text{Var}(X_1) \\ \text{Var}(X_2) \\ \vdots \\ \text{Var}(X_N) \end{bmatrix} \quad \text{but that does not account for the fact that these things covary}$$

we define the variance as a matrix!

$$\text{Var}(\underline{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_N) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots & \text{Cov}(X_2, X_N) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_N, X_1) & \text{Cov}(X_N, X_2) & \dots & \text{Var}(X_N) \end{pmatrix} \rightsquigarrow \text{symmetric}$$

$$\text{Var}(\underline{X}) = E[(\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T]$$

$$= E\left[\begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \\ \vdots \\ X_N - \mu_N \end{pmatrix} (X_1 - \mu_1, X_2 - \mu_2, \dots, X_N - \mu_N)\right]$$

$$= E\left[\begin{pmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) & \dots & (X_1 - \mu_1)(X_N - \mu_N) \\ \vdots & \vdots & \ddots & \vdots \\ (X_N - \mu_N)(X_1 - \mu_1) & (X_N - \mu_N)(X_2 - \mu_2) & \dots & (X_N - \mu_N)^2 \end{pmatrix}\right]$$

= Variance - covariance matrix

$$\text{Var}(A\underline{X}) = E[(A\underline{X} - A\underline{\mu})(A\underline{X} - A\underline{\mu})^T]$$

$$= E[A(\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T A^T]$$

$$= A E[(\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T] A^T$$

$$= A \text{Var}(\underline{X}) A^T$$

COVARIANCE AND CORRELATION

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\begin{matrix} X > \mu_X & + & + & + \\ X < \mu_X & - & - & + \\ X > \mu_X & + & - & - \\ X < \mu_X & - & + & - \end{matrix} \quad \begin{matrix} \nearrow \\ \searrow \\ \nearrow \\ \searrow \end{matrix}$$

$$\text{corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \rho \quad -1 \leq \rho \leq 1$$